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Productivity and Organization in Portuguese Firms

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1 Big picture

- **Question.** How does firm reorganization affect measured productivity? The paper asks whether changes in the number of management layers generate different effects on quantity-based productivity and revenue-based productivity.
- **Setting.** The paper studies Portuguese manufacturing firms from 1995 to 2005 using matched employer-employee records, balance-sheet information, and firm-product quantity and revenue data.
- **Core organizational margin.** A firm's organization is measured by its number of management layers. Firms can expand by adding a layer, contract by dropping a layer, or change scale without changing layers.
- **Main theory.** In a knowledge-hierarchy model, workers generate production problems and managers solve problems that lower layers cannot solve. Adding a management layer lets the firm operate efficiently at larger scale by reallocating knowledge and reducing average and marginal costs.
- **Quantity-productivity result.** When firms add layers, quantity-based productivity rises. In the dynamic panel specification, adding one layer is associated with about a 6% increase in quantity-based total factor productivity.
- **Revenue-productivity result.** When firms add layers, revenue-based productivity falls by roughly 3%. The reason is not that the firm becomes technologically worse. Rather, the firm expands output, moves down its demand curve, and charges lower prices.
- **Symmetric downsizing result.** When firms drop layers, the effects go in the opposite direction: quantity-based productivity falls, while prices and revenue-based productivity tend to rise.
- **Identification strategy.** The paper combines structural productivity estimation, OLS, dynamic panel instrumental variables, and a China-WTO textile quota shock that induced demand reductions and reorganization in affected textile and apparel firms.
- **Aggregate implication.** Reorganization is not just a side detail of growth. For firms that reorganize, the productivity change due to organization accounts for most, and sometimes more than all, of the observed mean quantity-productivity change.
- **Economic takeaway.** Measured productivity is partly endogenous to the firm's organizational choices. Revenue productivity can fall when real productive efficiency rises, so revenue-based productivity alone can misrepresent what happens when firms reorganize and change scale.

2 Incremental contribution of the paper

- **From productivity shocks to endogenous productivity responses.** Standard productivity analysis often treats productivity movements as shocks. This paper shows that an important component of measured productivity changes is the firm's endogenous organizational response to shocks.
- **A theory-consistent organization measure.** The paper uses occupational hierarchy data to measure the number of management layers, connecting observed organizational structure directly to the knowledge-based hierarchy theory of Garicano and Caliendo-Rossi-Hansberg.

- **Quantity versus revenue productivity.** A central contribution is separating quantity-based productivity from revenue-based productivity. The paper shows that these two measures can move in opposite directions when firms reorganize.
- **Unique Portuguese matched data.** The analysis uses a rare combination of worker-level occupation and wage data, firm balance sheets, and firm-product physical quantity data. This makes it possible to measure both organization and quantity-based productivity.
- **Structural productivity estimation with organization.** The paper adapts the MULAMA methodology to recover productivity, demand, and markup components while allowing changes in layers to enter the productivity process.
- **Causal evidence from multiple designs.** Beyond correlations, the paper uses dynamic panel instruments and a plausibly exogenous textile/apparel shock from China’s entry into the WTO to study reorganization caused by demand shocks.
- **Firm-level mechanism evidence.** The paper presents detailed reorganization examples showing how firms add or remove layers, change occupational composition, alter wages by layer, and change prices and quantities.
- **Pricing and pass-through implication.** By showing that reorganizing firms have unusually large price responses, the paper contributes to the cost pass-through literature and clarifies why conditional pass-through can differ from unconditional pass-through.
- **Development implication.** If firms in some environments cannot build larger hierarchies because monitoring, contracting, or managerial capacity is weak, they may fail to realize productivity gains from growth.

3 Data and measurement

3.1 Data sources

- **Employer-employee data.** The paper uses *Quadros de Pessoal*, a compulsory annual census of Portuguese firms with at least one worker. It records worker occupations, earnings, hours, firm identifiers, location, industry, employment, and sales.
- **Balance-sheet data.** The paper uses *Central de Balanços*, which provides firm-level balance-sheet information such as sales, material assets, material costs, and supplies and services.
- **Production data.** The paper uses *Inquérito Anual à Produção Industrial*, which records firm-product production values and physical quantities for manufacturing goods.
- **Time period and sector.** The data cover continental Portuguese manufacturing from 1995 to 2005. The matched employer-employee data are broad, while quantity data are available for a smaller sample of manufacturing firms.
- **Nested samples.** The larger sample combines employer-employee data with balance sheets and is used for organizational facts. The smaller firm-product sample adds physical quantity data and is used for the productivity analysis.

Interpretation: The paper’s strength is the combination of organizational data and quantity data. Most productivity studies can observe revenue and inputs but not physical quantities or detailed worker hierarchy.

3.2 Measuring management layers

- Workers are assigned to legally defined occupational categories based on tasks and skill requirements.
- The authors map these categories into four hierarchical occupations: occupation 0 for the lowest layer, occupation 1 for higher-skilled professionals, occupation 2 for middle management and supervisors, and occupation 3 for top executives.
- A firm with c occupied categories is said to have $L = c - 1$ management layers. Thus, the data contain firms with zero to three management layers.
- The empirical hierarchy is usually pyramidal: lower layers employ more hours and pay lower wages, while higher layers employ fewer workers and pay higher wages.
- Around 20% of firms add and/or drop occupations in a given year, implying a substantial amount of organizational restructuring.

Interpretation: The layer measure is simple but economically disciplined. It maps occupational structure into the hierarchy predicted by the theory, and the wage and employment patterns validate that it captures real organizational hierarchy.

3.3 Productivity measures

- **Quantity-based productivity** measures how efficiently a firm turns inputs into physical output. It removes the direct effect of output prices.
- **Revenue-based productivity** uses revenue rather than physical output and therefore combines technical efficiency with price and markup effects.
- The distinction is crucial because a firm that becomes more efficient may expand output and lower prices, causing revenue-based productivity to fall even as quantity-based productivity rises.
- The paper estimates productivity at the firm-product-year level, where products are aggregated at a two-digit unit-of-measure pair and multiproduct firms are split using product revenue shares.
- The production estimation is designed to account for productivity shocks, demand shocks, markups, and organizational changes.

Interpretation: The paper's main empirical insight depends on having both productivity measures. Without quantity data, the increase in real efficiency after adding layers could be hidden by the price decline.

3.4 Reorganization examples

- The paper gives several firm portraits to show the mechanism before the formal regressions.
- A growing aluminium cookware firm adds a management layer, increases quantity, lowers price, increases quantity-based productivity, and lowers revenue-based productivity.
- Textile and apparel firms hit by China-WTO quota removals drop layers, shrink output, focus on core tasks or products, raise prices, and show declining quantity productivity but rising revenue productivity.
- A footwear firm downsizes from three to two management layers, while a cork firm grows from one to two layers. Both examples show the predicted wage and productivity patterns.

Interpretation: The examples are not the main identification strategy, but they make the economic mechanism concrete: reorganizations involve occupational restructuring, scale changes, and price changes, not just relabeling of workers.

4 Theory and mechanisms

4.1 Knowledge hierarchy model

The paper builds on a knowledge-based hierarchy model. Workers use time and knowledge to produce and solve routine problems. If they cannot solve a problem, they pass it to a manager in the layer above. Higher layers have more knowledge and solve less frequent but more difficult problems.

A simplified production function is

$$Q_{it} = A_{it} O_{it}^{\alpha_O} M_{it}^{\alpha_M} K_{it}^{\gamma - \alpha_M - \alpha_O},$$

where O_{it} is the labor input generated by the hierarchy, M_{it} is material input, K_{it} is capital, and A_{it} is quantity-based productivity.

- The firm chooses how many workers to hire in each layer, how much knowledge they have, and how many management layers to use.
- A small firm may not need a complex hierarchy. A larger firm may find it efficient to add layers because managers economize on knowledge held by lower-level workers.
- The average cost curve is U-shaped for a given number of layers. As output expands, the firm eventually reaches a threshold where adding a new layer lowers average cost.

Interpretation: The key theoretical idea is that organization is lumpy. Firms do not continuously adjust hierarchy; they cross thresholds and then add or drop layers.

4.2 Adding layers

When a positive demand or productivity shock makes a firm expand enough to add a layer:

- the firm hires more workers in preexisting layers;
- workers in preexisting layers can be less knowledgeable and therefore earn lower wages, because the new upper layer handles more complex problems;
- the new organizational design is efficient at larger scale;
- average and marginal costs fall at the new scale;
- output rises and price falls as the firm moves down its demand curve.

Economic message: Adding a layer raises real productive efficiency but can lower revenue productivity through the price channel.

4.3 Dropping layers

When a negative demand shock makes a firm contract enough to drop a layer:

- the firm simplifies its hierarchy;

- it may focus on core tasks or core products;
- the quantity sold falls;
- the price tends to rise;
- quantity-based productivity falls but revenue-based productivity may rise.

Economic message: The model predicts asymmetric measurement effects: growth through reorganization can look bad in revenue productivity even when it raises true efficiency, and contraction can look good in revenue productivity even when real productive efficiency falls.

5 Models and empirical specifications

5.1 Production and timing

The firm's log production equation can be written as

$$q_{it} = \tilde{a}_{it} + \alpha_O \ln C(O_{it}^*; w_t) + \alpha_M m_{it} + (\gamma - \alpha_M - \alpha_O) k_{it},$$

where $C(O_{it}^*; w_t)$ is the total wage bill, m_{it} is material input, k_{it} is capital, and $\tilde{a}_{it}$ is measured quantity-based productivity.

- Capital is chosen before current-period productivity and demand shocks are realized.
- The firm observes some information about current shocks before choosing its number of management layers.
- Materials and labor inputs are chosen conditional on current shocks, capital, and layers.
- This timing makes layer changes potentially endogenous to current productivity and demand shocks.

Interpretation: The timing assumptions explain why the paper needs instruments for layer changes: firms do not reorganize randomly.

5.2 Demand, markups, and revenue productivity

The paper assumes generalized CES demand. In logs, revenue is

$$r_{it} = \left(1 - \frac{1}{\eta_{it}}\right) (\ell_{it} + q_{it}),$$

where ℓ_{it} is a demand shifter and η_{it} is the elasticity of demand.

Markups can be recovered from input expenditure shares:

$$\mu_{it} = \frac{\alpha_M}{s_{Mit}} = \frac{\alpha_O}{s_{Oit}}.$$

Revenue-based productivity is

$$a_{it}^* = r_{it} - \alpha_O \ln C(O_{it}^*; w_t) - \alpha_M m_{it} - (\gamma - \alpha_M - \alpha_O) k_{it}.$$

Interpretation: Revenue productivity equals quantity productivity plus price. Therefore, any price effect from reorganization mechanically matters for revenue TFP.

5.3 Productivity process

The paper models quantity-based productivity as

$$\tilde{a}_{it} = \alpha_i + d_t + \phi_a \tilde{a}_{it-1} + \phi_L \Delta L_{it} + v_{it}^a,$$

where ΔL_{it} is the change in the number of management layers.

- $\phi_L > 0$ means adding layers raises quantity-based productivity.
- For revenue-based productivity, the same reorganization term enters, but price and markup-related controls are required.
- The empirical analysis studies increasing sequences, decreasing sequences, constant-layer sequences, and all sequences pooled.

Interpretation: The coefficient on layer changes is the key empirical object. It measures how productivity changes when firms reorganize, conditional on lagged productivity and fixed effects.

6 Identification and empirical design

6.1 OLS specifications

The initial specifications estimate productivity equations by OLS with product or industry fixed effects, year effects, and lagged productivity. These results establish the baseline patterns:

- adding layers is positively related to quantity-based productivity;
- adding layers is negatively related to revenue-based productivity;
- dropping layers produces the reverse pattern.

Interpretation: OLS is useful descriptively but may be biased because firms change layers in response to current shocks.

6.2 Dynamic panel instrumental variables

The main dynamic panel estimates use an Arellano-Bover style system GMM estimator with firm-product-sequence fixed effects.

- The paper instruments layer changes using lagged demand, lagged markups, lagged layers, lagged revenue and quantity, lagged productivity, and capital variables implied by the model's timing.
- Firm-product-sequence fixed effects control for persistent heterogeneity in products and firms.
- The estimator also deals with the lagged dependent variable in a panel setting.

Interpretation: This design is stronger than OLS because it uses the model's timing to separate predicted reorganization from current productivity shocks.

6.3 China-WTO textile and apparel shock

The paper also studies the textile and apparel sector around China's entry into the WTO.

- Quota removals increased import competition for some textile and apparel products.

- The paper measures each firm-product’s exposure using the share of underlying HS6 products previously covered by quotas.
- This quota coverage variable instruments for layer changes in the 2000-2005 period.
- The shock is especially useful because it is a plausibly exogenous negative demand shock that induced downsizing and reorganization.

Interpretation: The textile exercise provides a clearer shock-based test of the mechanism: firms hit by quota removal dropped layers and experienced the productivity changes predicted by the theory.

6.4 Limitations of the design

- Layer changes are not randomly assigned. The causal interpretation depends on the validity of timing assumptions and instruments.
- The quantity-based productivity estimates require strong structural assumptions about demand, markups, and input choices.
- The sample is Portuguese manufacturing during 1995-2005, so external validity to services, modern digital firms, or other countries is not automatic.
- The layer measure is based on occupational categories; it captures hierarchy but not every kind of organizational change, such as plant creation, IT investment, or informal delegation.

Interpretation: The paper is persuasive because theory, firm examples, OLS, dynamic panel estimates, prices, and the textile shock all point in the same direction. Still, the exact magnitudes depend on the structural and instrument assumptions.

7 Tables and figures

7.1 Figures 1 and 2: Theory of layers and productivity jumps

Read:

- Figure 1 shows the average cost curve by number of layers. For each fixed number of layers, average cost is U-shaped, and the lower envelope determines when the firm reorganizes.
- The right panels of Figure 1 illustrate how employment and wages by layer change as the firm expands without changing layers versus expands by adding a layer.
- Figure 2 shows the main measurement prediction: when a firm adds a layer, quantity-based productivity jumps up, while revenue-based productivity jumps down.
- The downward jump in revenue productivity comes from the price response. A more efficient organization lets the firm expand output, which lowers price under downward-sloping demand.

Economic message: These figures contain the conceptual core of the paper. Organizational improvements can increase physical efficiency while reducing revenue productivity.

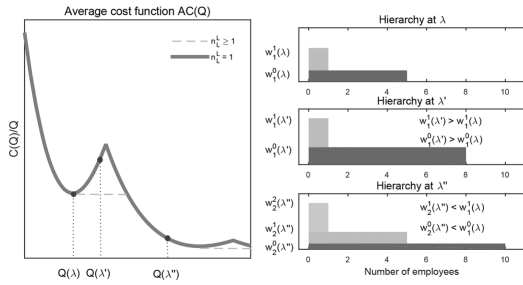


FIG. 1.—Average cost and organization.

(a) Figure 1: Average cost and organization

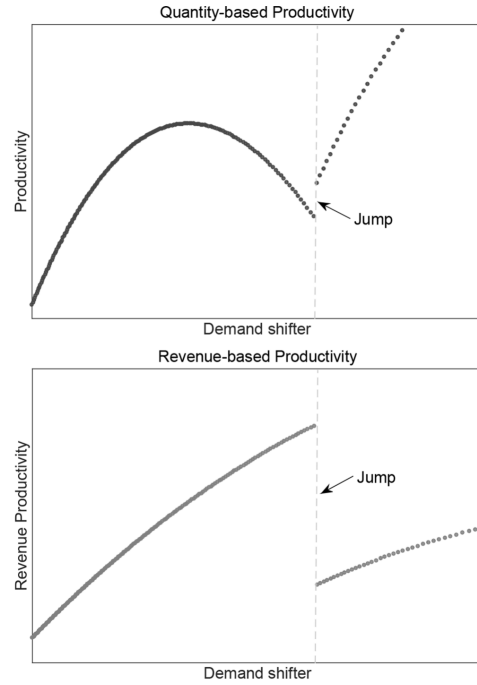


FIG. 2.—Changes to quantity- and revenue-based productivity as a firm adds layers.

(b) Figure 2: Quantity- and revenue-productivity changes when a firm adds layers

7.2 Figures 3 and 4: Firm example and sectoral patterns

Read:

- Figure 3 follows an aluminium cookware firm that adds a layer. Quantity rises sharply, price falls, and revenue rises more modestly.
- This pattern matches the theory: the firm becomes more efficient, expands, and lowers price.
- Figure 4 compares value added per worker by industry. For firms that do not change layers, growing sales are associated with higher value added per worker.
- For firms that change layers, the sign reverses: firms that increase layers often show lower value added per worker, while firms that decrease layers often show higher value added per worker.

Economic message: Figure 4 makes clear why organization matters for productivity measurement: reorganizing firms do not follow the same productivity-sales relationship as non-reorganizing firms.

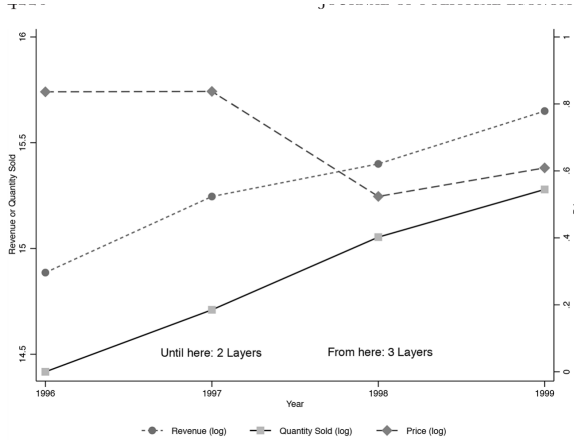


FIG. 3.—Example of a firm that adds layers: output, price, and revenue.

(a) Figure 3: Example of a firm that adds layers

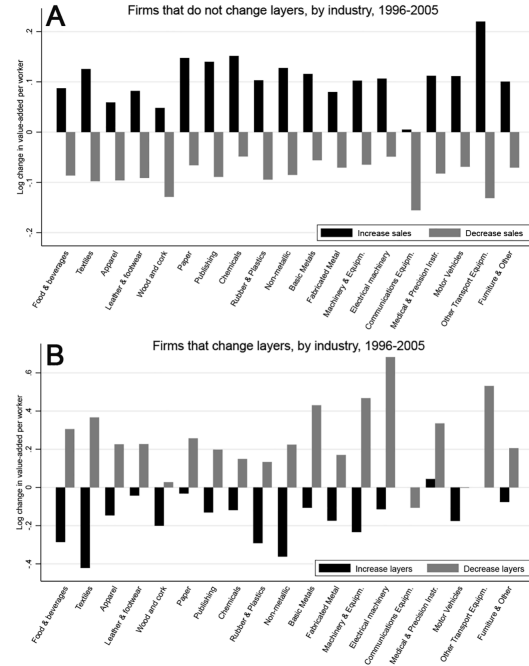


FIG. 4.—Reorganization and value added per worker.

(b) Figure 4: Reorganization and value added per worker

7.3 Tables 1 and 2: Textile and apparel firms hit by the China shock

Read:

- Table 1 shows a NACE 1772 textile/apparel firm that drops from three layers to two after the trade shock.
- The firm reduces employment from 37 to 10 workers, eliminates managers, narrows its task scope, and focuses on core production activities.
- Table 2 shows a NACE 1720 woven-fabrics firm that also drops from three layers to two.
- The firm reduces product scope by focusing on cotton fabrics and dropping synthetic fabrics.

Economic message: The tables show that dropping layers is a real organizational restructuring, not simply a small staffing adjustment. Firms simplify hierarchy, focus on core activities, and change production scope.

TABLE 1
TEXTILE AND APPAREL FIRM REORGANIZATION: NACE 1772 EXAMPLE

	Layer 0	Layer 1	Layer 2	Layer 3
Firm with three layers (2004):				
Managers			1	2
Technicians and associate professionals		1		
Clerks	2			
Crafts workers	15		4	
Plant and machine operators	11			
Elementary occupations	1			
Firm with two layers (2005):				
Managers				
Technicians and associate professionals		1	1	
Clerks		1		
Crafts workers	4			
Plant and machine operators	1		1	
Elementary occupations	1			

NOTE.—Occupations correspond to ISCO-88 one-digit major groups.

(a) Table 1: Textile and apparel firm reorganization, NACE 1772

TABLE 2
TEXTILE AND APPAREL FIRM REORGANIZATION: NACE 1720 EXAMPLE

	Layer 0	Layer 1	Layer 2	Layer 3
Firm with three layers (1999):				
Professionals				3
Technicians and associate professionals	1	5	2	
Clerks	14	10	2	
Crafts workers	17			
Plant and machine operators	71	2	1	
Elementary occupations	17			
Firm with two layers (2000):				
Professionals				2
Technicians and associate professionals	1	5	2	
Clerks	16	8	1	
Crafts workers	15			
Plant and machine operators	49	3	1	
Elementary occupations	17			

NOTE.—Occupations correspond to ISCO-88 one-digit major groups.

(b) Table 2: Textile and apparel firm reorganization, NACE 1720

7.4 Tables 3 and 4: Footwear and cork reorganization examples

Read:

- Table 3 shows a footwear firm that downsizes from three management layers to two. It removes managerial structure and reduces the number of crafts workers and elementary workers.
- The firm's quantity-based productivity falls, quantity falls, and price rises, consistent with the theory for dropping layers.
- Table 4 shows a cork firm that grows from one layer to two. It adds managers and slightly expands the worker layer.
- The cork example illustrates the positive-shock case: adding a layer increases quantity-based productivity and output while reducing price and revenue productivity.

Economic message: The four case-study tables together show that the theory applies to both downsizing and growth, and across several industries rather than only textiles.

TABLE 3
FOOTWEAR FIRM REORGANIZATION: NACE 19301352 EXAMPLE

	Layer 0	Layer 1	Layer 2	Layer 3
Firm with three layers (1998):				
Managers				1
Technicians and associate professionals		1		
Clerks		1		1
Crafts workers	56	1	1	
Elementary occupations	9			
Firm with two layers (1999):				
Managers				
Technicians and associate professionals		1		
Clerks		1		
Crafts workers	47		2	
Elementary occupations	7			

NOTE.—Occupations correspond to ISCO-88 one-digit major groups.

(a) Table 3: Footwear firm reorganization

TABLE 4
CORK FIRM REORGANIZATION: NACE 2052 EXAMPLE

	Layer 0	Layer 1	Layer 2	Layer 3
Firm with one layer (2004):				
Managers				
Clerks		1		
Crafts workers	8	1		
Firm with two layers (2005):				
Managers			2	
Clerks		1		
Crafts workers	9			

NOTE.—Occupations correspond to ISCO-88 one-digit major groups.

(b) Table 4: Cork firm reorganization

7.5 Tables 5 and 6: Quantity-based TFP regressions

Read:

- Table 5 reports OLS estimates for quantity-based TFP with product fixed effects. The coefficient on change in layers is positive and significant.
- In the pooled OLS specification, adding one layer is associated with about a 2.5% increase in quantity-based TFP.

- Table 6 reports dynamic panel estimates with firm-product-sequence fixed effects. The coefficient on change in layers is larger.
- In the pooled dynamic panel specification, adding one layer is associated with about a 6.2% increase in quantity-based TFP.

Economic message: The quantity-based evidence supports the core prediction: reorganizations that add layers make firms physically more productive at the new scale.

TABLE 5
QUANTITY-BASED TOTAL-FACTOR PRODUCTIVITY: ORDINARY LEAST SQUARES ESTIMATOR
WITH PRODUCT FIXED EFFECTS

	Increasing (1)	Decreasing (2)	Constant (3)	All (4)
Quantity-based productivity $t - 1$	.892*** (.014)	.875*** (.015)	.905*** (.013)	.895*** (.008)
Change in layers	.025** (.011)	.032*** (.012)		.025*** (.007)
Constant	-.161** (.063)	.112*** (.035)	.309** (.130)	.082** (.033)
Observations	4,141	2,829	3,031	10,001
Adjusted R^2	.779	.752	.801	.781

NOTE.—Year and industry dummies are included in the estimations. Firm-level-clustered standard errors are in parentheses.
** $p < .05$.
*** $p < .01$.

(a) Table 5: Quantity-based TFP, OLS

TABLE 6
QUANTITY-BASED TOTAL-FACTOR PRODUCTIVITY: DYNAMIC PANEL DATA ESTIMATOR
WITH FIRM-PRODUCT-SEQUENCE FIXED EFFECTS

	Increasing (1)	Decreasing (2)	Constant (3)	All (4)
Quantity-based productivity $t - 1$	.912*** (.012)	.880*** (.018)	.926*** (.014)	.910*** (.008)
Change in layers	.037** (.017)	.052** (.023)		.062*** (.016)
Constant	-.014 (.016)	.127 (.123)	.211*** (.042)	.116*** (.031)
Observations	4,141	2,829	3,031	10,001
Number of fixed effects	1,663	1,274	1,290	4,227
AR(2) test stat	.468	.117	2.443	1.980
p -value AR(2)	.640	.907	.015	.048

NOTE.—Year and industry dummies are included in the estimations. Firm-level-clustered standard errors are in parentheses.
** $p < .05$.
*** $p < .01$.

(b) Table 6: Quantity-based TFP, dynamic panel estimator

7.6 Tables 7 and 8: Revenue-based TFP regressions

Read:

- Table 7 reports OLS estimates for revenue-based TFP. The coefficient on change in layers is negative and significant.
- In the pooled OLS specification, adding one layer is associated with about a 2.7% decrease in revenue-based TFP.
- Table 8 reports dynamic panel estimates. The negative result remains after instrumenting and using firm-product-sequence fixed effects.
- In the pooled dynamic panel specification, adding one layer is associated with about a 2.5% decrease in revenue-based TFP.

Economic message: Tables 5-8 are the central empirical contrast. The same reorganization raises quantity productivity but lowers revenue productivity, exactly as the model predicts.

TABLE 7
REVENUE-BASED TOTAL-FACTOR PRODUCTIVITY: ORDINARY LEAST SQUARES ESTIMATOR
WITH PRODUCT FIXED EFFECTS

	Increasing (1)	Decreasing (2)	Constant (3)	All (4)
Revenue-based productivity $t - 1$	.925*** (.017)	.940*** (.021)	.955*** (.017)	.943*** (.011)
Change in layers	-.023*** (.006)	-.032*** (.006)	-.027*** (.003)	-.027*** (.003)
Demand $t - 1$	-.019*** (.004)	-.018*** (.003)	-.025*** (.003)	-.020*** (.003)
Price $t - 1$	.010* (.005)	.005 (.005)	.019*** (.007)	.012*** (.004)
Log markup	.484*** (.043)	.434*** (.044)	.553*** (.059)	.486*** (.036)
Capital	.004** (.002)	.003 (.003)	.005** (.002)	.004*** (.001)
Constant	.237*** (.046)	.095* (.051)	-.704*** (.153)	.707*** (.057)
Observations	4,141	2,829	3,031	10,001
Adjusted R^2	.851	.844	.882	.860

NOTE.—Year and industry dummies are included in the estimations. Firm-level-clustered standard errors are in parentheses.

* $p < .1$.
** $p < .05$.
*** $p < .01$.

(a) Table 7: Revenue-based TFP, OLS

TABLE 8
REVENUE-BASED TOTAL-FACTOR PRODUCTIVITY: DYNAMIC PANEL DATA ESTIMATOR
RESULTS WITH FIRM-PRODUCT-SEQUENCE FIXED EFFECTS

	Increasing (1)	Decreasing (2)	Constant (3)	All (4)
Revenue-based productivity $t - 1$	.935*** (.014)	.956*** (.019)	.967*** (.016)	.953*** (.009)
Change in layers	-.018** (.008)	-.035*** (.011)	-.025*** (.004)	-.025*** (.009)
Demand $t - 1$	-.006 (.003)	-.008*** (.002)	-.008* (.004)	-.006*** (.002)
Price $t - 1$	-.007 (.005)	-.011* (.006)	-.001 (.006)	-.006* (.003)
Log markup	.075 (.070)	.059 (.046)	.074 (.081)	.049 (.042)
Capital	.001 (.002)	.002 (.002)	.001 (.002)	.001 (.001)
Constant	-.027*** (.009)	.079 (.051)	.000 (.000)	-.014** (.006)
Observations	4,141	2,829	3,031	10,001
Number of fixed effects	1,663	1,274	1,290	4,227
AR(2) test stat	.043	1.352	1.548	1.805
pvalue AR(2)	.966	.177	.122	.071

NOTE.—Year and industry dummies are included in the estimations. Firm-level-clustered standard errors are in parentheses.

* $p < .1$.
** $p < .05$.
*** $p < .01$.

(b) Table 8: Revenue-based TFP, dynamic panel estimator

7.7 Tables 9 and 10: Textile IV and prices

Read:

- Table 9 focuses on textile and apparel firms exposed to quota removals after China's entry into the WTO.
- The IV estimate implies that a layer increase raises quantity-based productivity; equivalently, firms induced to drop layers by the shock experience a large fall in quantity-based productivity. The IV magnitude is about 14.7% in the quantity-productivity specification.
- The revenue-based effects in the textile IV are negative but imprecise.
- Table 10 estimates price responses. The coefficient on change in layers is negative, implying that adding layers reduces prices and dropping layers raises prices.

Economic message: The textile shock strengthens the causal interpretation, and the price regressions verify the mechanism behind the revenue-productivity decline.

TABLE 9
TEXTILE AND APPAREL: ORDINARY LEAST SQUARES (OLS)
AND INSTRUMENTAL VARIABLE (IV) ESTIMATES

	REVENUE-BASED TOTAL-FACTOR PRODUCTIVITY		QUANTITY-BASED TOTAL-FACTOR PRODUCTIVITY	
	OLS (1)	IV (2)	OLS (3)	IV (4)
Revenue-based productivity $t - 1$	.834*** (.040)	.827*** (.042)		
Quantity-based productivity $t - 1$			.865*** (.030)	.864*** (.030)
Change in layers	-.014 (.014)	-.026 (.018)	.085*** (.028)	.147** (.066)
Demand $t - 1$	-.011*** (.002)	-.008*** (.003)		
Price $t - 1$	.004 (.008)	.002 (.008)		
Log markup	.145*** (.033)	.097* (.058)		
Capital	.003* (.002)	.002 (.002)		
Constant	-.077** (.033)		.019 (.027)	
Observations	554	554	554	554
Adjusted R^2	.666	.660	.729	.725
Kleibergen-Paap statistic		32.50		42.03

NOTE.—Year dummies are included in the estimations. Firm-product-level-clustered standard errors are in parentheses.

* $p < .1$.
** $p < .05$.
*** $p < .01$.

(a) Table 9: Textile and apparel OLS/IV estimates

TABLE 10
PRICES: DYNAMIC PANEL DATA ESTIMATOR RESULTS
WITH FIRM-PRODUCT-SEQUENCE FIXED EFFECTS

	Increasing (1)	Decreasing (2)	Constant (3)	All (4)
Price $t - 1$	.878*** (.012)	.851*** (.021)	.905*** (.017)	.882*** (.009)
Change in layers	-.059*** (.015)	-.074*** (.021)		-.074*** (.015)
Constant	.162** (.067)	-.060 (.067)	-.383** (.152)	-.001 (.005)
Observations	4,141	2,829	3,031	10,001
Number of fixed effects	1,663	1,274	1,290	4,227
AR(2) test stat	.940	-.395	1.833	1.354
p -value AR(2)	.347	.693	.067	.176

NOTE.—Year and industry dummies are included in the estimations. Firm-level-clustered standard errors are in parentheses.

** $p < .05$.
*** $p < .01$.

(b) Table 10: Price responses to layer changes

7.8 Table 11: Aggregate quantity-productivity effects

Read:

- Table 11 decomposes quantity-TFP changes for firms that increase or reduce layers.
- For firms that increase layers, the mean overall quantity-TFP change is 0.066, while the component due to reorganization is 0.062.
- For firms that reduce layers, the mean overall quantity-TFP change is -0.019, while the reorganization component is -0.063.
- Because other shocks and mean reversion can offset or reinforce layer effects, the reorganization component can explain more than the total observed mean productivity change.

Economic message: Reorganization is quantitatively important. For reorganizing firms, layer changes account for most of the average productivity movement.

TABLE 11
CHANGE IN QUANTITY TOTAL-FACTOR PRODUCTIVITY DUE TO REORGANIZATION

Percentile (%)	FIRMS THAT INCREASE LAYERS		FIRMS THAT REDUCE LAYERS	
	Overall Change	Due to Reorganization	Overall Change	Due to Reorganization
10	-.483	.042	-.523	-.093
25	-.179	.051	-.272	-.062
50	.055	.056	-.034	-.062
75	.318	.062	.200	-.053
90	.673	.100	.517	-.047
Mean	.066	.062	-.019	-.063
Observations	810	810	465	465

Table 11: Change in quantity TFP due to reorganization

8 Short summary

- The paper studies how management-layer reorganization affects productivity in Portuguese manufacturing firms.

- Organization is measured using occupational hierarchy: firms have zero to three management layers.
- The theory predicts that adding a layer lowers average and marginal cost at larger scale.
- Adding layers should raise quantity-based productivity but lower revenue-based productivity through lower prices.
- Dropping layers should do the opposite: lower quantity-based productivity but raise prices and potentially revenue-based productivity.
- The empirical strategy uses matched employer-employee data, balance sheets, and firm-product quantity data.
- Quantity-based TFP rises by about 6% when firms add layers in the dynamic panel specification.
- Revenue-based TFP falls by about 3% when firms add layers.
- The China-WTO textile quota shock provides additional evidence that exogenous demand shocks lead firms to reorganize in ways that generate the predicted productivity effects.
- Price regressions show that adding layers lowers prices, directly supporting the mechanism behind the revenue-TFP decline.
- Aggregate decompositions show that reorganization explains most of the mean productivity change among firms that change layers.

9 Conclusion

9.1 Core conclusion

Caliendo, Mion, Opromolla, and Rossi-Hansberg show that firm productivity is shaped by organizational choices. When firms grow enough to add a management layer, they become more physically efficient and increase quantity-based productivity. At the same time, they expand output and lower prices, so revenue-based productivity falls. The reverse occurs when firms contract and drop layers.

9.2 Economic intuition

The intuition is that management layers are lumpy organizational capacity. A larger hierarchy is costly when the firm is small but efficient when the firm is large. Once a positive shock pushes the firm across the threshold for adding a layer, the new organization reduces costs at larger scale. This raises physical productivity, but the firm's larger output lowers prices, causing revenue productivity to move in the opposite direction.

9.3 Incremental contribution revisited

The paper advances the productivity literature by showing that measured productivity changes are not only exogenous shocks. They can be endogenous responses to shocks through firm reorganization. It also advances the organization literature by linking a concrete hierarchy measure - management layers from occupations - to structural productivity estimates and price responses.

9.4 Implications for productivity measurement

The paper warns against interpreting revenue-based productivity as pure technical efficiency. If a firm reorganizes, becomes more efficient, expands output, and lowers price, revenue productivity can fall even though real efficiency rises. This is especially important for studies of firm growth, exporting, trade shocks, and management practices.

9.5 Implications for firms and policy

The results imply that organizational capacity can be a barrier to growth. Firms that cannot build effective hierarchies may remain too small and fail to capture productivity gains. This point is relevant for developing economies where contracting, monitoring, or managerial talent constraints may make it difficult to expand managerial layers.

9.6 Limitations

The estimates depend on a structural productivity framework and on instruments motivated by timing assumptions. The paper studies manufacturing firms in Portugal from 1995 to 2005, so the magnitudes may not generalize to other sectors or countries. The layer measure captures formal occupational hierarchy but not all dimensions of organization, such as informal delegation, management quality, IT systems, or plant structure.

9.7 Takeaway

The enduring takeaway is that productivity is partly organizational. Adding managerial layers can make a firm truly more productive while making revenue productivity look worse. The paper therefore changes how we should read productivity statistics: part of measured productivity dynamics reflects how firms reorganize to exploit shocks and scale.